

Qwen-Audio

*Advancing Universal Audio Understanding via
Unified Large-Scale Audio-Language Models*

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Audio-language models II

Why this paper matters



The first audio-language model that is genuinely multi-everything

30+

tasks co-trained in a single
pre-training run

8

languages, across speech,
sound, music and song

2.0 / 4.2

WER on LibriSpeech test-clean
and test-other

0

task-specific fine-tuning on
any of the twelve benchmarks

● The claim

Scale audio-language pre-training to cover every kind of audio and every kind of task at once. The obstacle is interference: textual labels vary enormously across datasets in task focus, language, annotation granularity and text structure. Condition the decoder on a sequence of hierarchical tags and you get knowledge sharing where tasks are similar and separation where they are not.

● Why this is a different question from last session's

Pengi had eight hand-written templates. SALMONN had twelve instruction-tuning tasks and its problem was that they crowded out the language model's own abilities. Qwen-Audio has more than thirty tasks and its problem is that they crowd out each other — the same audio, asked for in two different formats, with no way for the model to tell which is wanted.

1

The interference problem

Why co-training thirty audio tasks does not just work



The one-to-many problem



Four axes along which audio task labels diverge

● Task focus

The same speech recording is a transcript for ASR, a language ID for LID, a gender label for SGC, an emotion label for ER, and a speaker identity for SV. One input, five incompatible targets.

● Language

Eight languages across the training mix, so even 'transcribe this' has eight different correct answers depending on which language the output should be in.

● Granularity of annotation

Some datasets have sentence-level timestamps, some word-level, most none at all. A model trained on all three has no way to know which is expected.

● Text structure

Some outputs are free-form captions, others are structured lists of attributes such as pitch and velocity. Different grammars entirely.

● The consequence the paper names

Simply mixing these diverse datasets cannot lead to mutual enhancement; instead, it introduces interference. The model is being asked to map one audio input to many mutually exclusive outputs with nothing in the input to disambiguate which one.

What everyone else did



Three prior approaches, and their limits

● Group similar tasks

Bundle captioning together, transcription together, and train each group separately or with a group marker. Coarse: it prevents the worst interference and forfeits sharing between groups.

● Assign a dataset ID

Give each dataset its own identifier token. Effective at separation, but every dataset becomes its own island — no knowledge sharing at all, which is the whole point of co-training.

● Whisper's task tokens

A template specifying task and condition information as special input tokens to the decoder: voice activity, language ID, sentence-level timestamps. The right idea — but Whisper only covers speech recognition and translation.

● What Qwen-Audio wants instead

Sharing where tasks are similar, separation where they are not, with the granularity chosen per axis rather than per dataset. That is what a hierarchy buys over a flat identifier.

2

The hierarchical tag framework

The paper's central contribution



Six tags, in order



Everything before the output text is a decision about what to produce

Position	Tag	Values	What it separates
1	Transcription tag	< startoftranscripts > · < startofanalysis >	Transcribing linguistic content from everything else
2	Audio language tag	One token per language, eight in total, or < unknown >	Which language is being spoken — unknown for music and sound
3	Task tag	transcribe · translate · caption · analysis · question-answer	The five task families; QA appends the question after the tag
4	Text language tag	One token per output language	Which language the output text should be in
5	Timestamps tag	< timestamps > · < notimestamps >	Word-level timestamp prediction, or none
6	Output instruction	Free text	The specific subtask and desired output format

The guiding principle, in the authors' words: maximise the sharing of knowledge among similar tasks through shared tags, while ensuring different tasks and output formats can be distinguished to avoid the one-to-many mapping problem.

The extra task: word-level timestamps



Speech recognition with word-level timestamp prediction — included deliberately, and ablated

● What it is

Unlike Whisper's sentence-level timestamps, the `<|timestamps|>` tag requires fine-grained word-level prediction.

The timestamps are interleaved with the words: a start-time token before each transcription token and an end-time token after it.

Twenty-one thousand hours of it — ten in English, eleven in Mandarin.

● Why the authors added it

Two stated purposes: to improve the model's ability to align audio signals with fine-grained timestamps, and to support grounding — finding when a name is mentioned, or whether a sound occurs at all.

The second is subtle: audio event detection becomes a subtask of timestamp prediction, since the absence of a timestamp implies the event did not occur.

● The surprising part

The claim is not just that SRWT helps timestamp prediction. It is that training on word-level timestamps improves plain speech recognition, and improves question answering about sound and music — tasks with no timestamps in them at all.

This is the one claim in the paper that gets a controlled ablation, and we look at the numbers in section 4.

3

Architecture, data and training

Whisper plus Qwen-7B, and an unusual freezing schedule



The architecture



Deliberately minimal — one encoder, one LLM, no connection module



- Audio is resampled to 16 kHz and converted to an 80-channel mel spectrogram with a 25 ms window and 10 ms hop, following Whisper.
- A pooling layer with stride two halves the sequence length, so each encoder output frame corresponds to about 40 ms of audio.
- SpecAugment is applied during training.
- Note what is absent: no Q-Former, no mapping network, no LoRA. The encoder output goes to the LLM directly, and the whole model is trained end to end in two stages.
- The rationale for Whisper as the encoder: although it was supervised for recognition and translation, its representations still contain rich information such as background noise, and can even be used to recover the original speech.

The training mix



Hours by task — and note what is small



- Emotion recognition is in the under-1,000-hour group, alongside overlapped speech recognition, diarisation, intent classification, slot filling, and every music task except captioning and genre.
- So the mix is dominated by transcription-shaped data: ASR plus SRWT alone is 51,000 hours.
- Recall the SALMONN lesson — deterministic, transcription-shaped tasks skew the model's prior over responses. The same risk applies here, in a mix that is even more transcription-heavy.
- The paper does not report a total, and the individual figures are given only as rounded hours.

Hold the emotion figure in mind. Under a thousand hours out of a mix exceeding a hundred thousand, and we will see the result it produces.

Two stages, with an unusual freezing schedule



Each stage trains what the other froze

● Stage 1 — multitask pre-training

Freeze the LLM. Train only the audio encoder, initialised from Whisper-large-v2.

500,000 steps, global batch 120, peak learning rate 5e-5 with cosine decay, AdamW, bfloat16.

Result: Qwen-Audio.

● Stage 2 — supervised fine-tuning

Freeze the audio encoder. Train only the LLM — fully, not with an adaptor.

8,000 steps, batch 128, learning rate 1e-5. Twenty thousand instruction samples, built by hand and with GPT-3.5, in ChatML format with multi-audio labelling.

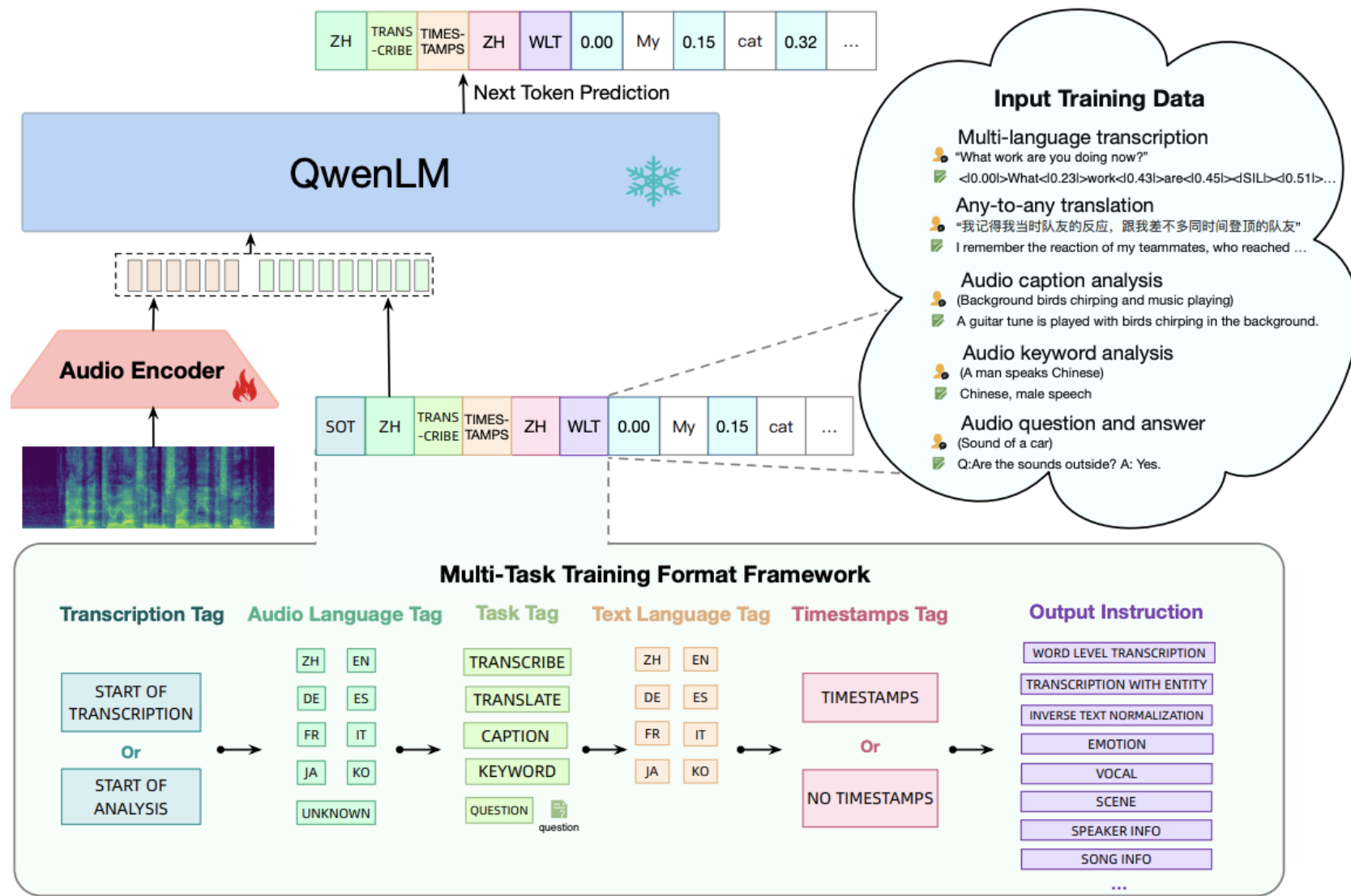
Result: Qwen-Audio-Chat.

● Why this schedule is worth noticing

Stage 1 pushes all the cross-modal adaptation into the audio encoder, leaving the language model's weights untouched — the same protection against catastrophic forgetting that Pengi, LTU and SALMONN all sought, achieved without a frozen-model adaptor.

Stage 2 then fully fine-tunes the LLM on only twenty thousand samples, which is small enough that the risk is contained. It is a defensible design and the paper never argues for it.

Two stages, with an unusual freezing schedule



4

Results

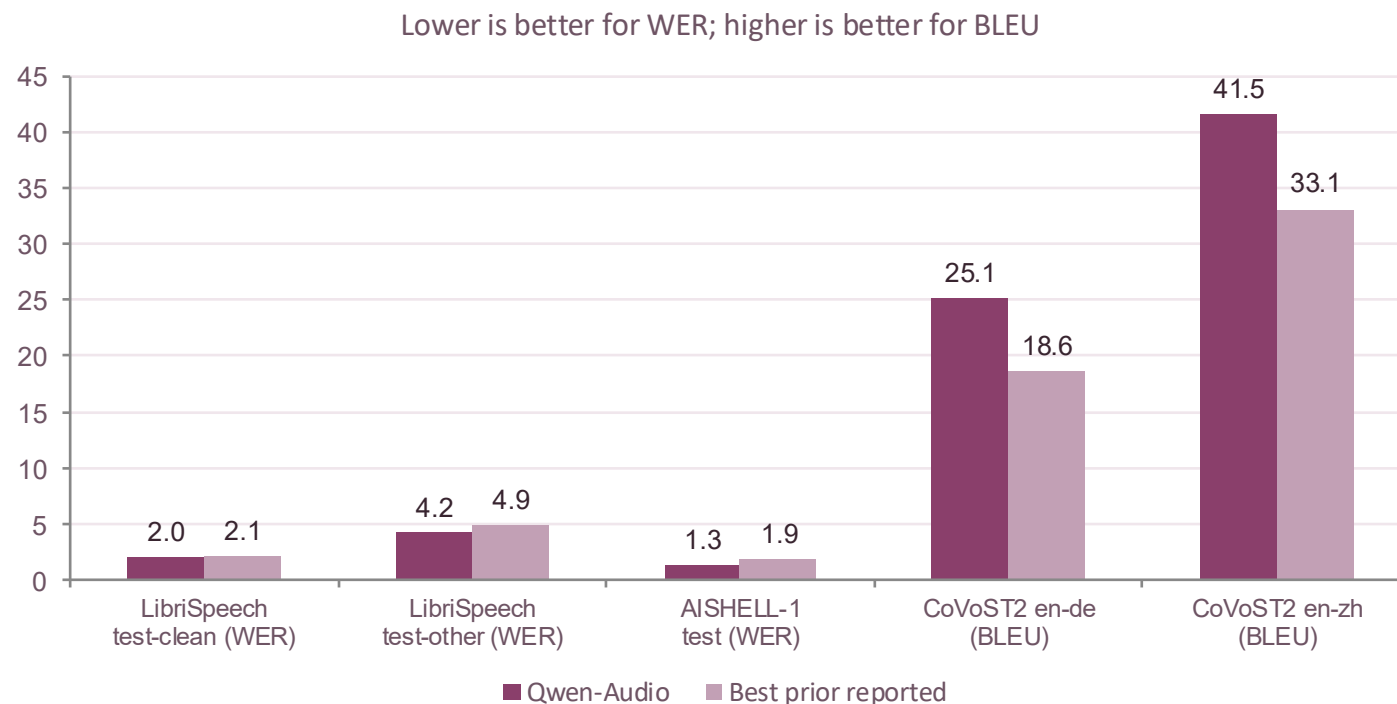
Twelve datasets, no task-specific fine-tuning



Speech recognition and translation



The strongest results in the paper



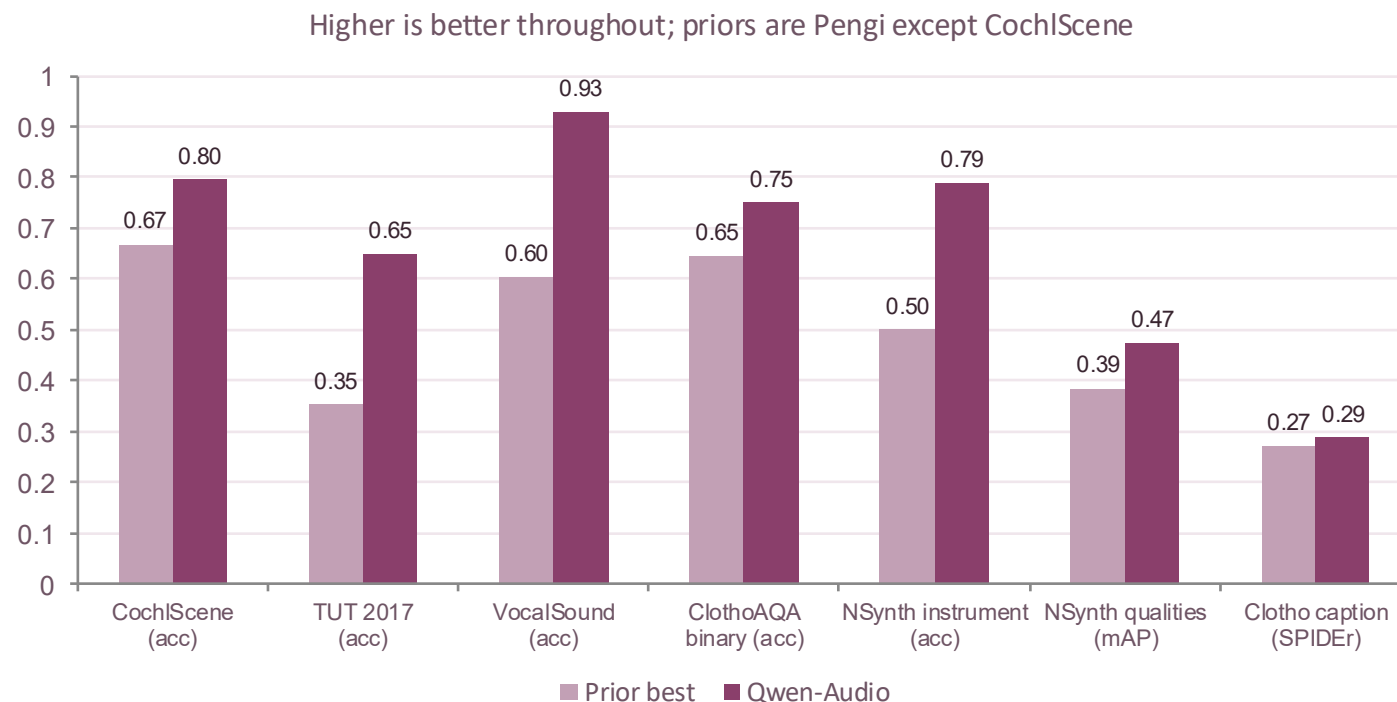
- The prior values are SALMONN for LibriSpeech and the translation directions, and MMSpeech-large for AISHELL-1.
- State of the art on AISHELL-1 dev and test, at 1.2 and 1.3 word error rate.
- Translation gains are large and consistent across all seven directions: de-en 33.9, es-en 39.7, fr-en 38.5, it-en 36.0, zh-en 15.7.
- Word-level timestamp prediction reaches 51.5 ms average alignment score, better than a forced aligner given the ground-truth transcript at 60.3 ms.

A single model doing this while also captioning music and analysing note velocity is the paper's real argument for the tag framework.

Sound, music and question answering



Against Pengi and CLAP from the previous session



- The VocalSound jump is the largest: 0.60 for Pengi, 0.93 for Qwen-Audio.
- TUT 2017 acoustic scenes nearly doubles, from 0.35 to 0.65.
- NSynth instrument classification goes from 0.50 to 0.79 — a music task, from a model whose encoder was trained for speech recognition.
- Captioning is the narrowest margin: 0.288 against Pengi's 0.271 SPIDER on Clotho.

Pengi's own strength — sound and music classification — is where Qwen-Audio's margins are widest. Scale and task breadth beat a purpose-built sound-event encoder.

The emotion result



One line in the table, and worth a slide

● What is reported

Speech emotion recognition on MELD, accuracy 0.557, against WavLM-large at 0.542.

A 1.5-point improvement over a strong self-supervised baseline.

● What we know about MELD

Seven classes, heavily imbalanced toward neutral, drawn from the Friends TV series.

From the CLEP-DG session: on MELD, weighted accuracy around 50 can sit alongside unweighted accuracy around 23, on a task where chance is about 14%.

● What is not reported

Only accuracy. No unweighted accuracy, no per-class breakdown, no F1.

And emotion is one of the under-1,000-hour tasks in a mix exceeding a hundred thousand hours.

● How to read it

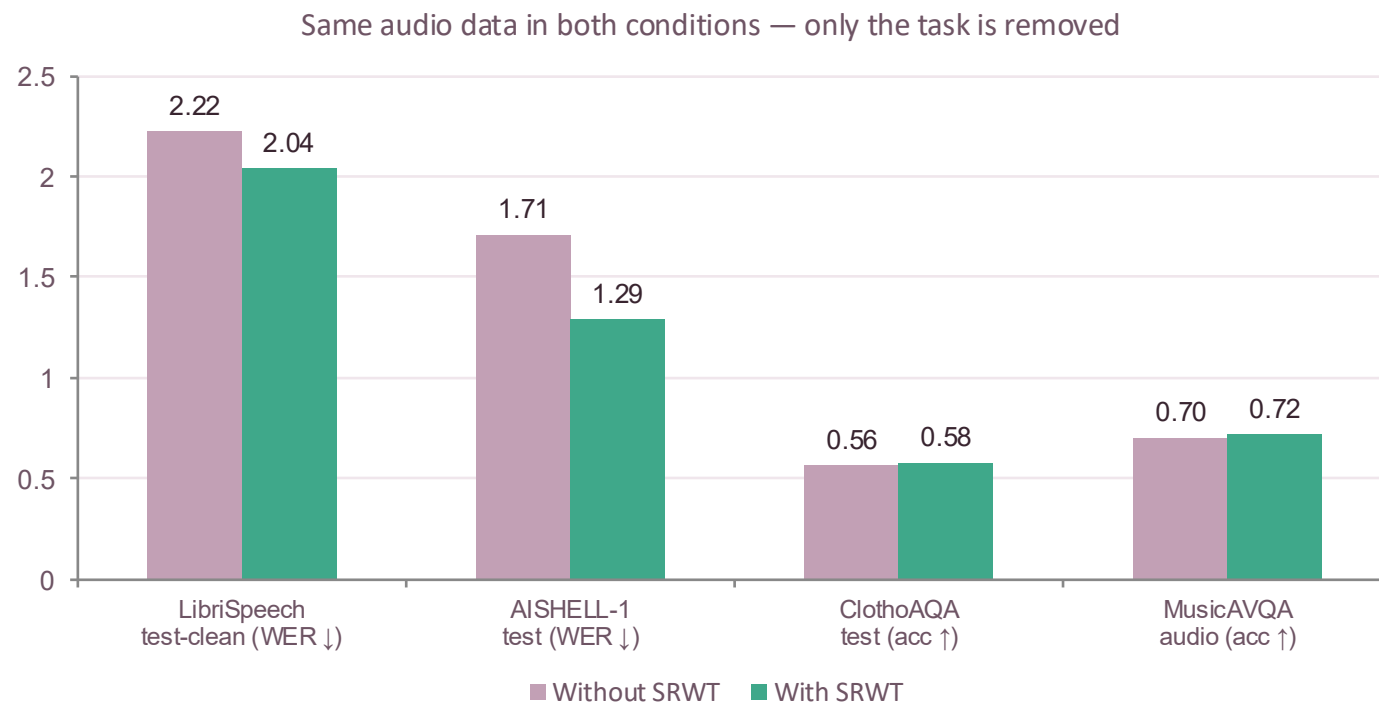
The number is not wrong and the comparison is fair — WavLM-large is measured the same way. But a single weighted accuracy on a skewed seven-class set carries very little information about the rare emotions, which is exactly the AffectNet and CLEP-DG lesson.

The honest summary of this session's two papers on paralinguistics: emotion is present in the task mix, reported with one number, and never examined.

The one ablation



Does word-level timestamp training help tasks that have no timestamps?



- The control is clean: removing SRWT does not change which audio the model sees, because SRWT and ASR share the same datasets.
- Mandarin ASR improves most — 1.71 to 1.29 word error rate, a relative gain of 25%.
- And the transfer claim holds: sound question answering improves by 1.5 points and music question answering by 1.8, on tasks with no timestamps in them.
- This is the only ablation in the paper, and it supports the secondary contribution rather than the headline one.

A finer-grained alignment task improves coarser ones. The same pattern we saw in HuBERT, where a harder pretext task produced better representations.

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Appraisal

What is demonstrated, and what is asserted



The missing experiment



The headline contribution is never tested

● What the paper claims

That conditioning on a sequence of hierarchical tags encourages knowledge sharing and avoids interference — and that this is what makes co-training thirty tasks work.

The claim appears in the title, the abstract, the contributions list and the conclusion.

● What would test it

Train the same model with a flat dataset ID instead of the hierarchy. Or with no tags at all. Or with a partial hierarchy — task tags but no language tags.

Each is a single training run on the same data, and each would say how much the structure is worth.

● Why this matters for what we can conclude

Qwen-Audio is bigger than its comparators, trained on more data, for more steps, with a stronger language model and a stronger encoder. Every one of those could explain the results.

So the leaderboard establishes that the system works. It does not establish that the hierarchy is why. Compare SALMONN, which ran four controls on its central claim, or LTU after the break, which ablates its curriculum directly.

Critical appraisal



What the design buys, and what it costs

● Strengths

Genuine breadth: speech, sound, music and song, thirty-plus tasks, eight languages, one model, no task-specific fine-tuning at evaluation.

State-of-the-art speech recognition and translation while also doing music note analysis — the combination is the achievement.

The tag hierarchy is a clean idea, well described, and directly generalises Whisper's task tokens.

SRWT is a real contribution with a properly controlled ablation, and the transfer result is non-obvious.

Evaluation datasets rigorously excluded from training to avoid leakage, and stated as such.

Code, checkpoints and demo released.

● Limitations

The tag framework — the headline claim — is never ablated.

Qwen-Audio-Chat is evaluated only through illustrative cases. No quantitative measure of dialogue quality, instruction following, or whether full LLM fine-tuning cost language ability.

Baselines are best-available numbers from different systems, not controlled comparisons.

Emotion reported as a single weighted accuracy on a skewed seven-class set.

Training data is not released, and several dataset sizes are given only as 'under 1,000 hours'.

A technical report, not peer reviewed — which shows in the absence of negative results.

What came next



And the bridge to the second half

● Qwen2-Audio

The direct successor replaced the hierarchical tags with natural-language prompts, and added a preference-optimisation stage — evidence that the tag framework was a transitional idea rather than a durable one.

● The Qwen-Audio backbone

Widely adopted as a starting point for audio-language work, and the lineage runs into the omni-modal Qwen models that combine audio, vision and text.

● Multilingual audio-LLMs

Eight languages in a single co-trained model set an expectation that these systems should not be English-only — a standard the earlier models in this series did not meet.

● What it leaves open

The model perceives extremely well and is never asked to reason. There is no evaluation of explanation, inference, or open-ended questioning beyond a fixed answer set — which is exactly what the second paper is about.

Take-aways



Four things to carry into the second half

- 1 Structured conditioning solves one-to-many**

When one input maps to many valid outputs, the disambiguating signal has to be in the input. A graded hierarchy shares more than a flat dataset ID.
- 2 Fine-grained alignment tasks transfer**

Word-level timestamp training improved speech recognition and question answering about music. Harder alignment, better representations — the HuBERT lesson again.
- 3 Check whether the headline claim was tested**

Thirty tasks, twelve benchmarks, one ablation — and the ablation is on the secondary contribution. Demonstrating a system is not testing a hypothesis.
- 4 Breadth and depth are different achievements**

This model perceives across speech, sound and music better than anything before it, and is never asked to explain, infer, or refuse.

Questions for discussion



1

The tag framework is never ablated, and every other difference from the baselines could explain the results. What is the cheapest experiment that would isolate it — and would you trust the paper's conclusion without it?

2

Word-level timestamp training improves question answering about music. What is the mechanism, and does it predict which other auxiliary tasks would transfer?

3

The successor, Qwen2-Audio, replaced hierarchical tags with natural-language prompts. What does that say about structured conditioning versus instruction following as interfaces?

4

Emotion is under 1,000 hours in a mix exceeding 100,000, and is reported as one weighted accuracy on MELD. Across all four audio-LLMs in these two sessions, why does paralinguistics keep coming out thin?

5

Qwen-Audio-Chat is half the contribution and is evaluated only by example. What minimum quantitative evaluation should a dialogue model in a paper like this be expected to report?